Second Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Chenxi Wu (TR 1:00 PM)	Niclas Technau (MWF 1:20 PM)	Ander Aguirre (TR 9:30 AM)	Sarah Tammen (MWF 12:05 PM)

- **There are 6 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	18	
2	10	
3	20	
4	19	
5	15	
6	18	
Total:	100	

1. In any parts of the following problem for which you deem it relevant, you may leave answers in terms of the function Φ (the CDF of a standard normal random variable) or its inverse.

   (a) (5 points) The dean of a certain university wants to estimate the proportion of students that are left-handed. He plans to send a survey to a random sample of the student body. If the dean wants to obtain a 95% confidence interval for the proportion of students that are left-handed and he wants an interval of width ε, then how many students should he survey? Give an answer of the form $n \geq f(\varepsilon)$.

   (b) (5 points) Suppose the dean's survey finds that the proportion of students that are left-handed is $p = 0.10$. Suppose that Professor Bob is teaching a 160 person Calculus class in a room with 20 left-handed desks. Use an appropriate approximation to estimate the probability that there will be more than 20 left-handed students in the class. (Assume that the hand preferences of different students are independent.)

(c) (5 points) Suppose that 0.1% (i.e. 1/1000) of the students in the university are ambidextrous. Suppose that the university admits a freshman class of $10,000$ students. Use an appropriate approximation to estimate the probability that exactly two of the new freshmen are ambidextrous.

(d) (3 points) We sample 40 students with replacement from the student body. Let X_1 be the number of times we choose a left-handed student, let X_2 be the number of times we choose an ambidextrous student, and let X_3 be the number of times we choose a right-handed student. Under the assumptions of parts (b)-(c), what distribution does the vector (X_1, X_2, X_3) have? (For full credit, you must specify all parameters.)

2. (10 points) Let X be a random variable with mean -4, and variance 16.
Find numbers $a, b > 0$ so that the random variable $Y = aX + b$ has mean 0 and variance 1.

3. Let $X \sim \text{Unif}([-1, 3])$. Compute the probability density functions of two the random variables:

(a) (10 points) $Y = X^3$

(b) (10 points) $Y = |X|$

4. In this problem we consider the moment generating function of a discrete random variable X.

(a) (11 points) Suppose the moment generating function of X is

$$M_X(t) = \frac{1}{8} + \frac{3}{8}e^t + \frac{3}{8}e^{2t} + \frac{1}{8}e^{3t}.$$

Derive the probability mass function of X.

(b) (7 points) Use the derivatives of M_X to find the variance $\mathrm{Var}(X)$.

(c) (1 point) Can you identify the distribution of X with parameters?

5. Let (X, Y) be uniformly distributed on the triangle with vertices $(0, 0)$, $(1, 1)$, and $(1, 2)$.

(a) (12 points) Find the marginals of the coordinates f_Y and f_X.

(b) (3 points) Are the two coordinates independent or not?

6. Let $\lambda > 0$. Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(2\lambda)$ be independent exponentials.

(a) (9 points) Compute the probability $P(X > 2Y)$. Write the answer in terms of λ.

(b) (9 points) Let $V = \min(X, Y)$. Derive a formula for the probability density function f_V of V.

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

x	$\Phi(x)$
-2.0	0.022
-1.9	0.028
-1.8	0.035
-1.7	0.044
-1.6	0.054
-1.5	0.066
-1.4	0.080
-1.3	0.096
-1.2	0.115
-1.1	0.135
-1.0	0.158
-0.9	0.184
-0.8	0.211
-0.7	0.242
-0.6	0.274
-0.5	0.308
-0.4	0.344
-0.3	0.382
-0.2	0.420
-0.1	0.460
0.0	0.500
0.1	0.539
0.2	0.579
0.3	0.617
0.4	0.655
0.5	0.691
0.6	0.725
0.7	0.758
0.8	0.788
0.9	0.815
1.0	0.841
1.1	0.864
1.2	0.884
1.3	0.903
1.4	0.919
1.5	0.933
1.6	0.945
1.7	0.955
1.8	0.964
1.9	0.971
2.0	0.977